

**Symbolic versus statistical reasoning:**

The (Symbolic) methods basically represent uncertainty belief as being

- True,
- False, *or*
- Neither True nor False.

Some methods also had problems with

- Incomplete Knowledge
- Contradictions in the knowledge.

Statistical methods provide a method for representing beliefs that are not certain (or uncertain) but for which there may be some supporting (or contradictory) evidence.

Statistical methods offer advantages in two broad scenarios:

**Genuine Randomness**

-- Card games are a good example. We may not be able to predict any outcomes with certainty but we have knowledge about the likelihood of certain items (*e.g.* like being dealt an ace) and we can exploit this.

**Exceptions**

-- Symbolic methods can represent this. However if the number of exceptions is large such system tend to break down. Many common sense and expert reasoning tasks for example. Statistical techniques can *summarise* large exceptions without resorting enumeration.

**Uncertain Knowledge:**

Let action  $A_t$  = leave for airport  $t$  minutes before flight. Will  $A_t$  get me there on time?

Problems:

1. Partial observability (road state, other drivers' plans, etc.)
2. Noisy sensors (radio traffic reports)
3. Uncertainty in action outcomes (flat tyre, etc.)
4. Complexity of modeling and predicting traffic

Hence a purely logical approach either

1. Risks falsehood: “ $A_{25}$  will get me there on time” or

2. Leads to conclusions that are too weak for decision making: “A<sub>25</sub> will get me there on time if there's no accident on the bridge and it doesn't rain and my tires remain intact etc etc.”

A<sub>1440</sub> might reasonably be said to get me there on time but I'd have to stay overnight in the airport...

### **Handling Uncertainty:**

Instead of providing all condition it can express with degree of beliefs in the relevant sentences.

Example:

Say we have a rule

*if toothache then problem is cavity*

But not all patients have toothaches because of cavities (although perhaps most do)

So we could set up rules like

*if toothache and not(gum disease) and not(filling) and .....then problem is cavity*

This gets very complicated! a better method would be to say

*if toothache then problem is cavity with probability 0.8*

Given the available evidence,

*A<sub>25</sub> will get me there on time with probability 0.04*

A most important tool for dealing with degree of beliefs is probability theory, which assigns to each sentence a numerical degree of belief between 0 & 1.

### **Making decisions under uncertainty:**

Suppose I believe the following:

$$P(A_{25} \text{ gets me there on time} | \dots) = 0.04$$

$$P(A_{90} \text{ gets me there on time} | \dots) = 0.70$$

$$P(A_{120} \text{ gets me there on time} | \dots) = 0.95$$

$$P(A_{1440} \text{ gets me there on time} | \dots) = 0.9999$$

Which action to choose?

- Depends on my preferences for missing flight vs. length of wait at airport, etc. Utility theory is used to represent and infer preferences

Decision theory = utility theory + probability theory

*The rational decision depends on both the relative importance of various goals and the likelihood that, and degree to which, they will be achieved.*

### Basic Statistical methods – Probability:

The basic approach statistical methods adopt to deal with uncertainty is via the axioms of probability:

- Probabilities are (real) numbers in the range 0 to 1.
- A probability of  $P(A) = 0$  indicates total uncertainty in  $A$ ,  $P(A) = 1$  total certainty and values in between some degree of (un)certainly.
- Probabilities can be calculated in a number of ways.

Very Simply

Probability = (number of desired outcomes) / (total number of outcomes)

So given a pack of playing cards the probability of being dealt an ace from a full normal deck is 4 (the number of aces) / 52 (number of cards in deck) which is  $1/13$ . Similarly the probability of being dealt a spade suit is  $13 / 52 = 1/4$ .

Conditional probability,  $P(A|B)$ , indicates the probability of of event  $A$  given that we know event  $B$  has occurred.

The aim of a **probabilistic logic** (or **probability logic**) is to combine the capacity of probability theory to handle uncertainty with the capacity of deductive logic to exploit structure. The result is a richer and more expressive formalism with a broad range of possible application areas. Probabilistic logic is a natural extension of traditional logic truth tables: the results they define are derived through probabilistic expressions instead. The difficulty with probabilistic logics is that they tend to multiply the computational complexities of their probabilistic and logical components.

### Random Variables:

In probability theory and statistics, a **random variable** (or **stochastic variable**) is a way of assigning a value (often a real number) to each possible outcome of a random event. These values might represent the possible outcomes of an experiment, or the potential values of a quantity whose value is uncertain (e.g., as a result of incomplete information or imprecise measurements.) Intuitively, a random variable can be thought of as a quantity whose value is not fixed, but which can take on different values; normally, a probability distribution is used to describe the probability of different values occurring. Random variables are usually real-valued, but one can consider arbitrary types such as boolean values, complex numbers, vectors, matrices, sequences, trees, sets, shapes, manifolds and functions. The term *random element* is used to encompass all such related concepts.

For example: There are two possible outcomes for a coin toss: heads, or tails. The possible outcomes for one fair coin toss can be described using the following random variable:

$$X = \begin{cases} \text{head,} \\ \text{tail.} \end{cases}$$

and if the coin is equally likely to land on either side then it has a probability mass function given by:

$$\rho_X(x) = \begin{cases} \frac{1}{2}, & \text{if } x = \text{head,} \\ \frac{1}{2}, & \text{if } x = \text{tail.} \end{cases}$$

**Example:** A simple world consisting of two random variables:

**Cavity**– a Boolean variable that refers to whether my lower left wisdom tooth has a cavity

**Toothache**- a Boolean variable that refers to whether I have a toothache or not

We use the single capital letters to represent unknown random variables

P induces a probability distribution for any random variables X.

Each RV has a domain of values that it can take it, e. g. domain of *Cavity* is {true, false}

**RVs domain are: Boolean, Discrete and Continuous**

**Atomic Event:**

An **atomic event** is a complete specification of the state of the world about which the agent is uncertain.

**Example:**

In the above world with two random variables (Cavity and Toothache) there are only four distinct atomic events, one being:

Cavity = false, Toothache = true

Which are the other three atomic events?

**Propositions:**

Think of a proposition as the event (set of sample points) where the proposition is true

Given Boolean random variables A and B:

event  $\alpha$  = set of sample points where  $A(\omega) = \text{true}$

event  $\neg\alpha$  = set of sample points where  $A(\omega) = \text{false}$

event  $a \wedge b$  = points where  $A(\omega) = \text{true}$  and  $B(\omega) = \text{true}$

Often in AI applications, the sample points are defined by the values of a set of random variables, i.e., the sample space is the Cartesian product of the ranges of the variables.

With Boolean variables, sample point = propositional logic model  
e.g.,  $A = \text{true}$ ,  $B = \text{false}$ , or  $a \wedge \neg b$ .

Proposition = disjunction of atomic events in which it is true  
e.g.,  $(a \vee b) \equiv (\neg a \wedge b) \vee (a \wedge \neg b) \vee (a \wedge b)$   
 $P(a \vee b) = P(\neg a \wedge b) + P(a \wedge \neg b) + P(a \wedge b)$

*Propositional or Boolean random variables*

e.g., *Cavity*(do I have a cavity?)

*Discrete random variables (finite or infinite)*

e.g., *Weather* is one of (*sunny*, *rain*, *cloudy*, *snow*)

*Weather* = *rain* is a proposition

Values must be exhaustive and mutually exclusive

*Continuous random variables (bounded or unbounded)*

e.g., *Temp* = 21.6, also allow, e.g., *Temp* < 22.0.

### Prior Probability:

The prior or unconditional probability associated with a proposition is the degree of belief accorded to it in the absence of any other information.

#### Example:

$P(\text{Weather} = \text{sunny}) = 0.72$ ,  $P(\text{Weather} = \text{rain}) = 0.1$ ,  $P(\text{Weather} = \text{cloudy}) = 0.08$ ,  
 $P(\text{Weather} = \text{snow}) = 0.1$

**Probability distribution** gives values for all possible assignments:

$P(\text{Weather}) = (0.72, 0.1, 0.08, 0.1)$

**Joint probability distribution** for a set of r.v.s gives the probability of every atomic event on those r.v.s (i.e., every sample point)

$P(\text{Weather}, \text{Cavity})$  = a  $4 \times 2$  matrix of values.

| Weather=     | sunny | rain | cloudy | snow |
|--------------|-------|------|--------|------|
| Cavity=true  | 0.144 | 0.02 | 0.016  | 0.02 |
| Cavity=false | 0.576 | 0.08 | 0.064  | 0.08 |

Every question about a domain can be answered by the joint distribution because every event is a sum of sample points.

**Conditional Probability:**

The conditional probability “ $P(a|b)$ ” is the probability of “a” given that all we know is “b”.

Example:  $P(\text{cavity}|\text{toothache}) = 0.8$  means if a patient have toothache and no other information is yet available, then the probability of patient's having the cavity is 0.8.

Definition of conditional probability:

$$P(a|b) = P(a \wedge b)/P(b) \text{ if } P(b) \neq 0$$

Product rule gives an alternative formulation:

$$P(a \wedge b) = P(a|b)P(b) = P(b|a)P(a)$$

**Inference using full joint probability distribution:**

We use the full joint distribution as the knowledge base from which answers to all questions may be derived. The probability of a proposition is equal to the sum of the probabilities of the atomic events in which it holds.

$$P(a) = \sum P(e_i)$$

Therefore, given a full joint distribution that specifies the probabilities of all the atomic events, one can compute the probability of any proposition.

**Full Joint probability distribution : an example**

We consider the following domain consisting of three Boolean variables: Toothache, Cavity, and Catch (the dentist's nasty steel probe catches in my tooth).

The full joint distribution is the following 2x2x2 table:

|               | toothache |              | $\neg$ toothache |              |
|---------------|-----------|--------------|------------------|--------------|
|               | catch     | $\neg$ catch | catch            | $\neg$ catch |
| Cavity        | 0.108     | 0.012        | 0.072            | 0.008        |
| $\neg$ cavity | 0.016     | 0.064        | 0.144            | 0.576        |

The probability of any proposition can be computed from the probabilities in the table. The probabilities in the joint distribution must sum to 1.

Each cell represents an atomic event and these are all the possible atomic events.

$$\begin{aligned}
 P(\text{cavity or toothache}) &= P(\text{cavity, toothache, catch}) + P(\text{cavity, toothache, } \neg\text{catch}) + \\
 &\quad P(\text{cavity, } \neg\text{toothache, catch}) + P(\text{cavity, } \neg\text{toothache, } \neg\text{catch}) + \\
 &\quad P(\neg\text{cavity, toothache, catch}) + P(\neg\text{cavity, toothache, } \neg\text{catch}) \\
 &= 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28
 \end{aligned}$$

We simply identify those atomic events in which the proposition is true and add up their probabilities

**Marginalization or summing out:**

Distribution over **Y** can be obtained by summing out all the other variables from any joint distribution containing **Y**. This process is called marginalization.

$$P(Y) = \sum P(Y, z)$$

Examples:

$$P(\text{cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2$$

$$P(\neg \text{Toothache}) = 0.072 + 0.008 + 0.144 + 0.576 = 0.8$$

$$P(\text{Cavity}, \neg \text{Toothache}) = 0.072 + 0.008 = 0.08$$

$$P(Y) = \sum P(Y, z)$$

$$P(Y, z) = P(Y|z)P(z)$$

Therefore, for any set of variables **Y** and **Z**:

$$P(Y) = \sum P(Y|z)P(z) - \text{This rule is the conditioning rule}$$

**Calculating Conditional Probability:**

$$\begin{aligned} P(\neg \text{cavity} \mid \text{Toothache}) &= P(\neg \text{cavity} \wedge \text{Toothache}) / P(\text{Toothache}) \\ &= (0.016 + 0.064) / (0.108 + 0.012 + 0.016 + 0.064) \\ &= 0.4 \end{aligned}$$

Again let's calculate

$$\begin{aligned} P(\text{cavity} \mid \text{Toothache}) &= P(\text{cavity} \wedge \text{Toothache}) / P(\text{Toothache}) \\ &= (0.108 + 0.012) / (0.108 + 0.012 + 0.016 + 0.064) \\ &= 0.6 \end{aligned}$$

Notice that in above two calculations the term  $1 / P(\text{Toothache})$  remain constant no matter which value of cavity is calculated. This constant term is called normalization constant for the distribution  $P(\text{cavity} \mid \text{Toothache})$ , ensuring that it adds up to 1.

**Independence:**

A and B are independent iff

$$P(A|B) = P(A) \text{ or } P(B|A) = P(B) \text{ or } P(A, B) = P(A)P(B)$$

Example:

$$P(\text{Toothache}, \text{Catch}, \text{Cavity}, \text{Weather}) = P(\text{Toothache}, \text{Catch}, \text{Cavity})P(\text{Weather})$$

Here weather is independent of other three variables.

**Bayes' Rule (Theorem) :**

$$P(b|a) = \frac{P(a|b) * P(b)}{P(a)}$$

Proof of bays rule:

We know that:

$$P(a|b) = P(a \wedge b) / P(b)$$

$$P(a \wedge b) = P(a|b) P(b) \dots \dots \dots (1)$$

Similarly

$$P(b|a) = P(a \wedge b) / P(a)$$

$$P(a \wedge b) = P(b|a) P(a) \dots \dots \dots (2)$$

Equating 1 and 2

$$P(a|b) P(b) = P(b|a) P(a)$$

$$\text{i.e. } P(b|a) = P(a|b) P(b) / P(a)$$

***Why is the Bayes' rule is useful in practice?***

Bayes' rule is useful in practice because there are many cases where we have good probability estimates for three of the four probabilities involved, and therefore can compute the fourth one.

Useful for assessing diagnostic probability from causal probability:

$$P(\text{Cause}|\text{Effect}) = \frac{P(\text{Effect}|\text{Cause})P(\text{Cause})}{P(\text{Effect})}$$

Diagnostic knowledge is often more fragile than causal knowledge.

**Example of Bayes' rule:**

A doctor knows that the disease meningitis causes the patient to have a stiff neck 50% of the time. The doctor also knows that the probability that a patient has meningitis is 1/50,000, and the probability that any patient has a stiff neck is 1/20.

Find the probability that a patient with a stiff neck has meningitis.



Here, we are given;

$$\begin{aligned}p(s|m) &= 0.5 \\p(m) &= 1/50000 \\p(s) &= 1/20\end{aligned}$$

Now using Bayes' rule;

$$P(m|s) = P(s|m)P(m)/P(s) = (0.5 * 1/50000)/(1/20) = 0.0002$$

### Uses of Bayes' Theorem :

In doing an expert task, such as medical diagnosis, the goal is to determine identifications (diseases) given observations (symptoms). Bayes' Theorem provides such a relationship.

$$P(A | B) = P(B | A) * P(A) / P(B)$$

Suppose:  $A$  = Patient has measles,  $B$  = has a rash

$$\text{Then: } P(\text{measles}/\text{rash}) = P(\text{rash}/\text{measles}) * P(\text{measles}) / P(\text{rash})$$

The desired diagnostic relationship on the left can be calculated based on the known statistical quantities on the right.

### Bayesian networks:

- A data structure to represent the dependencies among variables and to give a concise specification of any full joint probability distribution.
- Also called *belief networks* or *probabilistic network* or *casual network* or *knowledge map*.

The basic idea is:

- Knowledge in the world is *modular* -- most events are conditionally independent of most other events.
- Adopt a model that can use a more local representation to allow interactions between events that *only* affect each other.
- Some events may only be *unidirectional* others may be *bidirectional* -- make a distinction between these in model.
- Events may be causal and thus get chained together in a network.

### A Bayesian network is a directed acyclic graph which consists of:

- A set of random variables which makes up the nodes of the network.
- A set of directed links (arrows) connecting pairs of nodes. If there is an arrow from node  $X$  to node  $Y$ ,  $X$  is said to be a parent of  $Y$ .

- Each node  $X_i$  has a conditional probability distribution  $P(X_i | \text{Parents}(X_i))$  that quantifies the effect of the parents on the node.

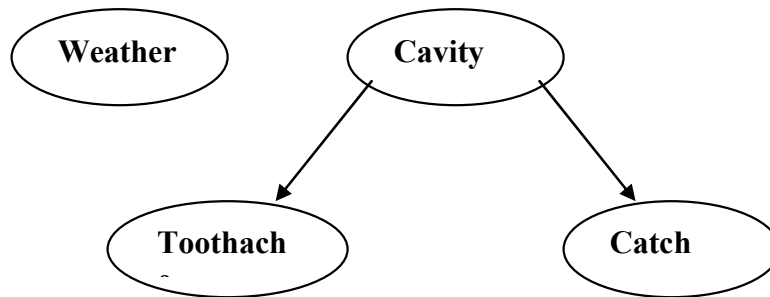
**Intuitions:**

- A Bayesian network models our incomplete understanding of the causal relationships from an application domain.
- A node represents some state of affairs or event.
- **A link from X to Y means that X has a direct influence on Y.**

**Implementation:**

- *A Bayesian Network is a directed acyclic graph:*
  - A graph where the directions are links which indicate dependencies that exist between nodes.
  - Nodes represent propositions about events or events themselves.
  - Conditional probabilities quantify the strength of dependencies.

Our existing simple world of variables *toothache*, *cavity*, *catch* & *weather* is represented as:



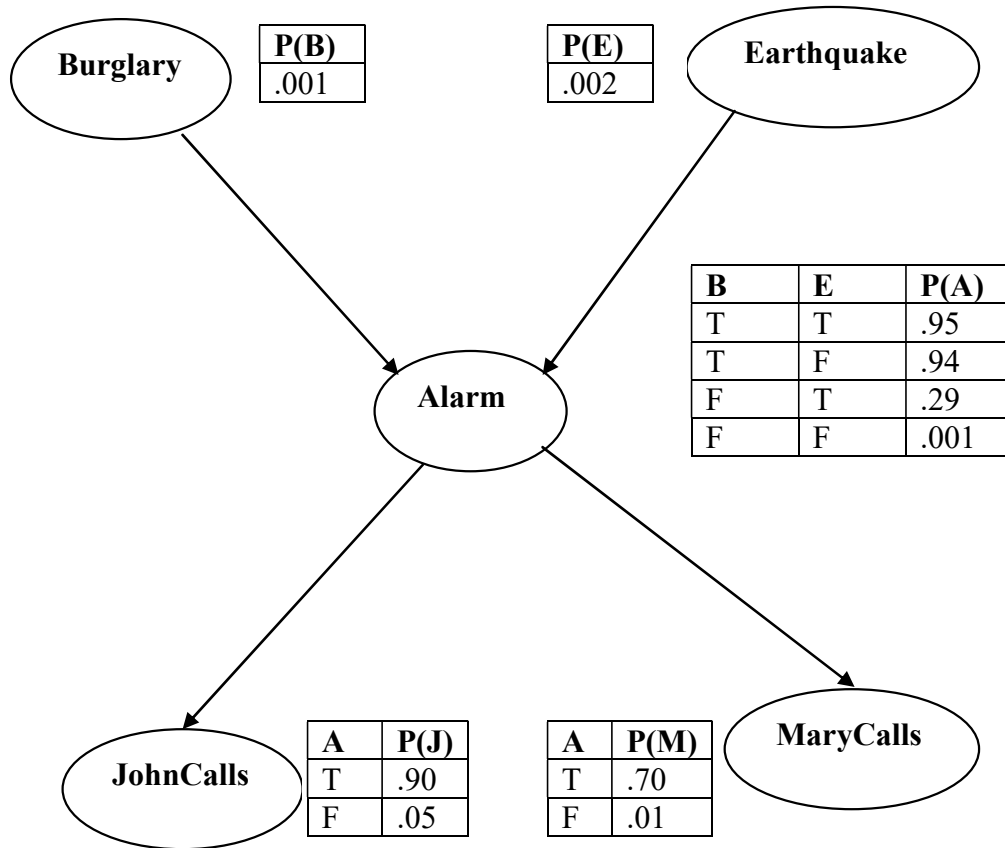
*Weather* is independent of the other variables

**Example:****Sample Domain:**

You have a burglar alarm installed in your home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes. You also have two neighbors, John and Mary, who have promised to call you at work when they hear the alarm. John always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too. Mary, on the other hand, likes rather loud music and sometimes misses the alarm altogether.

We would like have to estimate the probability of a burglary with given evidence who has or has not call.

**Variables:**Burglary, Earthquake, Alarm, JohnCalls, MaryCalls



The probabilities associated with the nodes reflect our representation of the causal relationships.

A Bayesian network provides a complete description of the domain in the sense that one can compute the probability of any state of the world (represented as a particular assignment to each variable).

**Example:** What is the probability that the alarm has sounded, but neither burglary nor an earthquake has occurred, and both John and Mary call?

$$P(j, m, a, \neg b, \neg e) = P(j|a) P(m|a) P(a, \neg b, \neg e) P(\neg b) P(\neg e)$$

$$= 0.90 * 0.70 * 0.001 * 0.999 * 0.998 = 0.00062$$

**Consider the following example:**

- The probability,  $P(S_1)$  that my car won't start.
- If my car won't start then it is likely that
  - The battery is flat or
  - The starting motor is broken.

In order to decide whether to fix the car myself or send it to the garage I make the following decision:

- If the headlights do not work then the battery is likely to be flat so i fix it myself.
- If the starting motor is defective then send car to garage.
- If battery and starting motor both gone send car to garage.

The Bayesian network to represent this is as follows:

